

Lab 1 – GradPace Product Description

GradPace – Team Copper

Carlycia Mathis

Old Dominion University

CS411W

Professor Sarah Hosni

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1 Introduction

Workforce expectations are rapidly changing, creating uncertainty for college students entering the job market. According to the World Economic Forum, 63% of employers cite skill gaps as their top barrier to growth, while IDC surveys indicate that 66% of enterprises are reducing entry-level hiring because of artificial intelligence integration. Furthermore, workplace automation studies show that 91% of organizational reports emphasize that automation is actively changing or eliminating standard job functions, reflecting significant structural shifts in the modern job market. This increases disconnect across higher education, career services, and the entry-level job market since 48% of recent graduates already feel underqualified for entry-level roles, and 50% to 60% end up underemployed (Cengage Group, 2025).

Universities use a tracking system for academic progress, but actual career preparation is fragmented. Existing tools only address isolated needs coursework tracking, basic job boards, or networking, leaving a significant gap in career readiness support. Students end up dealing with fragmented data, delayed gap detection, completely reactive planning, and really limited guidance. GradPace addresses this gap by providing a unified, real-time career readiness roadmap.

2 Product Description

GradPace is a career readiness platform designed for students, alumni, and industry professionals that prioritizes proactive professional preparation over traditional job board aggregation. Users

input academic and professional profiles, and the system generates a personalized roadmap that aligns current individual progress with real-time industry standards. Core capabilities include early warning gap detection and industry readiness benchmarking, which identify skill deficiencies prior to market entry by integrating live industry data and location-based professional networking. Through this automated synchronization, the platform ensures users align with modern workforce requirements from initial system onboarding.

Building on this foundation, the platform tracks how students transition from the classroom to the modern workforce by eliminating reactive planning and delayed gap detection. Instead of juggling isolated tools such as basic course trackers, generic job boards, and disconnected networking applications, users gain access to a centralized ecosystem that continuously aligns their current coursework and skills with actual industry standards. Ultimately, the objective is to empower every student and recent graduate with actionable insights and local professional connections, ensuring they enter the job market fully prepared, highly competitive, and confident in their career trajectory.

2.1 Key Product Features and Capabilities

GradPace functions as a comprehensive career readiness ecosystem that bridges academic progress and professional employment through secure account management, streamlined subscription controls, and dedicated student input interfaces. What makes the platform truly unique and innovative is its departure from traditional, reactive job boards by integrating live industry data to constantly sync student skill sets with real job skill requirements. Rather than tracking achievements, the platform accomplishes its mission through an automated, personalized roadmap and a visual dashboard that maps out a student's custom professional path.

This directly solves the problem of fragmented career planning and delayed gap detection by utilizing specialized features early warning gap detection to spot missing qualifications and industry readiness benchmarking to measure true workforce preparedness. Platform incorporates GPS tracking for local events and direct alumni connections, ensuring users gain actionable guidance and localized networking opportunities

2.2 Major Components (Hardware/Software)

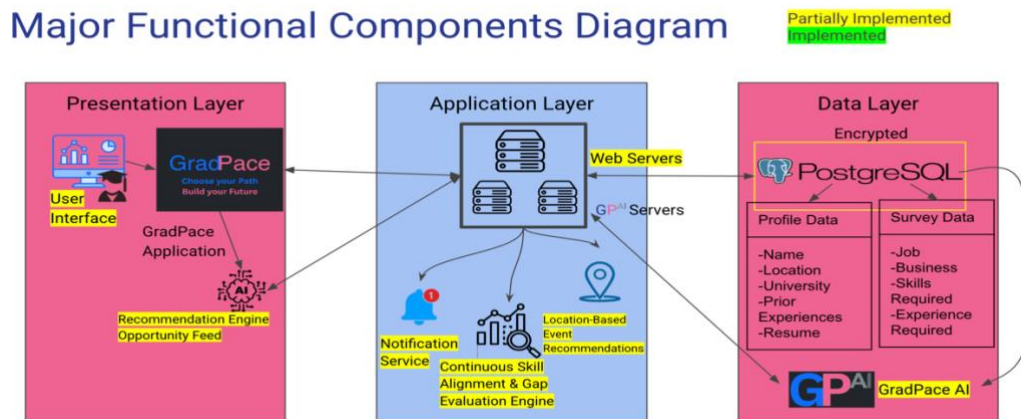


Figure 1 - MFCD

Operating as a web-based platform, the system requires minimal client-side hardware, consisting of a standard computer and a reliable internet connection to access cloud services. Following the structural design of the Major Functional Component Diagram (MFCD), the software architecture is organized into three primary structural layers. The Presentation Layer, the Application Layer, and the Data Layer. The Presentation Layer houses the user interface and the recommendation engine opportunity feed. The Application Layer handles core operational logic through web servers, GPAI servers, notification services, continuous skill alignment and gap evaluation engines, and location-based event recommendations. Finally, the Data Layer

incorporates an encrypted PostgreSQL database to securely manage profile data, survey data, and GradPace AI integration.

3 Identification of Case Study

The primary objective of this case study is to introduce the reader to a practical example of how the GradPace platform is utilized by a sample beta test group. This case study serves as a crucial bridge between the real-world problem statements and the process of finding a career after obtaining a degree. Furthermore, it outlines the specific target users, the roles they play, the features being exercised, and the tangible yield those features provide, all while protecting the capability roadmap.

3.1 Case Study Participants and Roles

The case study focuses on early undergraduate students navigating transition from academia to the modern workforce. The key role involved is Alex, a computer science undergraduate in his second semester of college. Alex acts as the primary student user who is eager to build professional development experience outside of standard classwork but feels overwhelmed by traditional, reactive job hunting and fragmented career resources

3.2 Features and Use Case Scenario

Alex inputs his transcripts, current work interests, and location information into GradPace overcoming his uncertainty regarding external professional development. The application processes this data through its core backend architecture and API integrations by exercising key

capabilities such as opportunity finding features, custom roadmap generation, transcript importing, and personalized goal setting based on user background. As a result, the platform instantly generates a personalized roadmap that outlines specific skill milestones, project opportunities, and local professional connections. Transitioning away from reactive job hunting to receive proactive guidance through a structured, actionable plan that aligns his ongoing academic choices with actual industry standards .

Looking toward future implementations, the platform is positioned to expand beyond individual student use into a broader institutional ecosystem. Future iterations will incorporate features tailored for universities, educators, and alumni networks, enabling academic advisors to monitor cohort-level skill gaps, allowing faculty to align curriculum benchmarks with live industry data, and facilitating mentorship connections between current students and established alumni.

Table 1

Table of Comparison Between RWP and Prototype

Feature	Prototype	Real World Product
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Account Management	PC display	Live user database
Personal data input	Form entry	Student profile
roadmap	4-year plan but can depend on students overall goals/plan	AI driven personalized paths
Gap Detection	checklist	Real time Alerts & Benchmarking
Local networking events	Simulated event list	GPS-based events

Table 1 illustrates the architectural comparison between the initial GradPace prototype and the target Real World Product (RWP). The table outlines the pragmatic development strategies employed to validate core platform mechanics such as utilizing PC displays, manual form entries, static checklists, and simulated event lists without compromising long-term platform scalability. By demonstrating system feasibility through these foundational interfaces, the implementation establishes a robust technical framework for the eventual deployment of advanced components, including live user databases, AI-driven personalized paths, real-time benchmarking alerts, and GPS-based event tracking.

3.3 Prototype Development Challenges

Knowledge Gaps may have API integration issues. GradPace may have issues pulling live data for our industry maps and getting live information from company to assist for the featured roadmaps. Database Migration, making sure queries are working correctly for custom roadmaps and making sure the data flows. Real-Time data parsing takes actual logic to adjust the students milestone checklist. The designing process may cause some havoc because GradPace is supposed to be a simple design, but we must find a way to grab the user's attention as well with

sleek styling. Other challenges may occur , but I know GradPace will be a fundamental developing tool.

4 Glossary

API (Application Programming Interface): A set of rules that allow different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Gap Detection: The automated process of analyzing a user's current academic transcripts and skill inventory against live industry standards to identify missing qualifications or professional competencies.

Industry Readiness Benchmarking: The systematic evaluation of a user's profile and skill set against current industry standards and employer expectations to measure professional preparedness.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allow users to interact with it.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

Career Roadmap: A personalized, actionable development plan generated by the platform that outlines specific skill milestones, academic choices, and project opportunities tailored to a user's background.

Career Readiness: The attainment and demonstration of requisite competencies and skills necessary to successfully transition from higher education into the professional workforce.

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